Bug Report — HW4: I2C Protocol

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One critical defect found by analysing I2C traffic in PulseView together with the UART log.

Environment: ESP32 DevKit with peripheral MPU6050 / 0x68 (SDA 21, SCL 22), Wokwi, PulseView

Summary

ID	Title	Severity
Bug	I2C communication does not recover after FAULT	Critical

Bug — I2C communication does not recover after FAULT

Environment	ESP32 DevKit with peripheral MPU6050 / 0x68 (SDA 21, SCL 22), Wokwi, PulseView
Preconditions	Simulator is started
Steps	1. Start the system with FAULT OFF. 2. Observe normal I2C communication with the MPU6050. 3. Verify that UART periodically displays: • [DATA] Temperature: 24.00 C • [SYSTEM] Main loop alive 4. Turn FAULT ON to simulate loss of the I2C peripheral. 5. Observe the I2C traffic in PulseView and the UART output. 6. Turn FAULT OFF without restarting the system. 7. Observe whether I2C communication and the main loop recover.
Expected	After the I2C device becomes available again, the system should recover communication automatically, resume receiving temperature data, and continue the main loop without requiring a restart.
Actual	After FAULT ON, the I2C transaction ends with a NACK: Address Write 69 → NACK → STOP. UART reports: requestFrom(): i2cWriteReadNonStop returned Error -1 After this error, the [DATA] Temperature and [SYSTEM] Main loop alive messages stop. After FAULT OFF, the system does not recover automatically. No successful I2C communication or temperature data is observed. A restart is required to restore normal operation.
Trace evidence	Transactions before the fault complete successfully with ACKs. The first observed failed transaction contains: Address Write 69 → NACK → STOP. No recovery ACK is observed after FAULT OFF.
Firmware log evidence	Normal operation: [DATA] Temperature: 24.00 C Normal operation: [SYSTEM] Main loop alive Error: requestFrom(): i2cWriteReadNonStop returned Error -1
Reproducibility	5/5
Severity	Critical

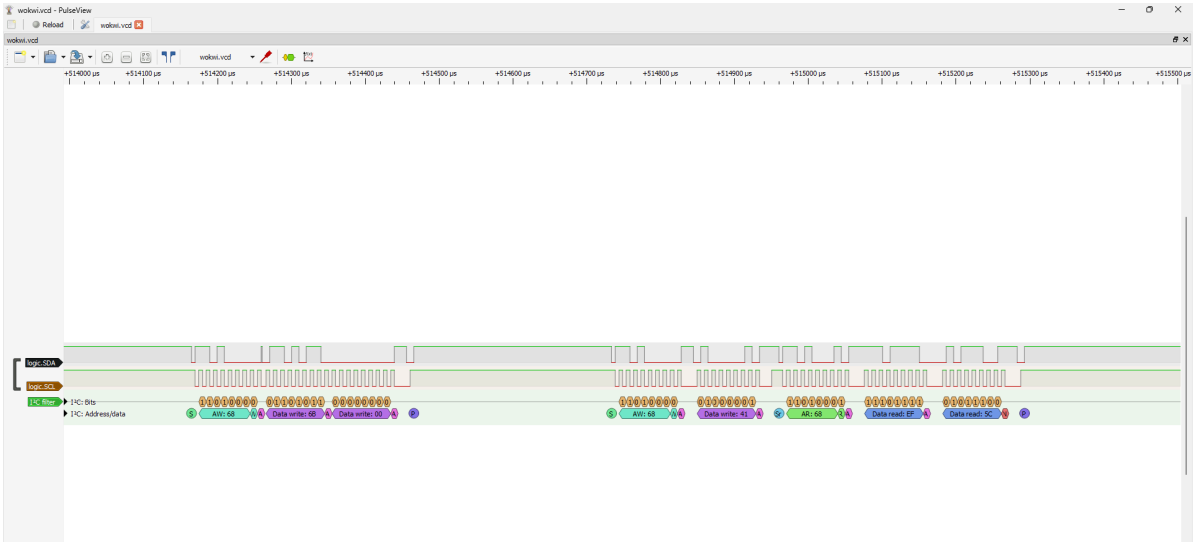


Figure: PulseView trace — 1st transaction

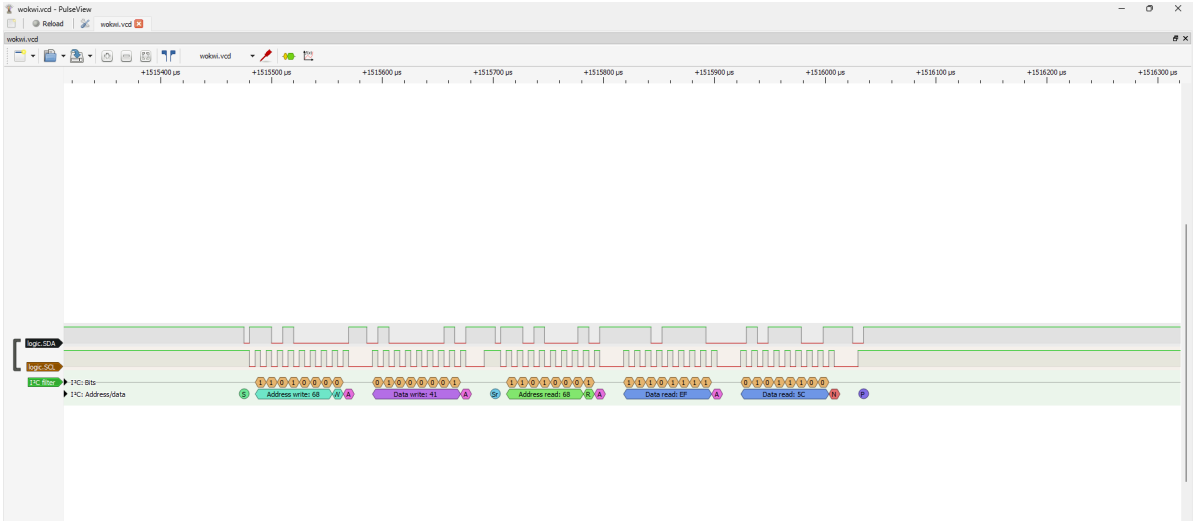


Figure: PulseView trace — 2nd–10th transactions

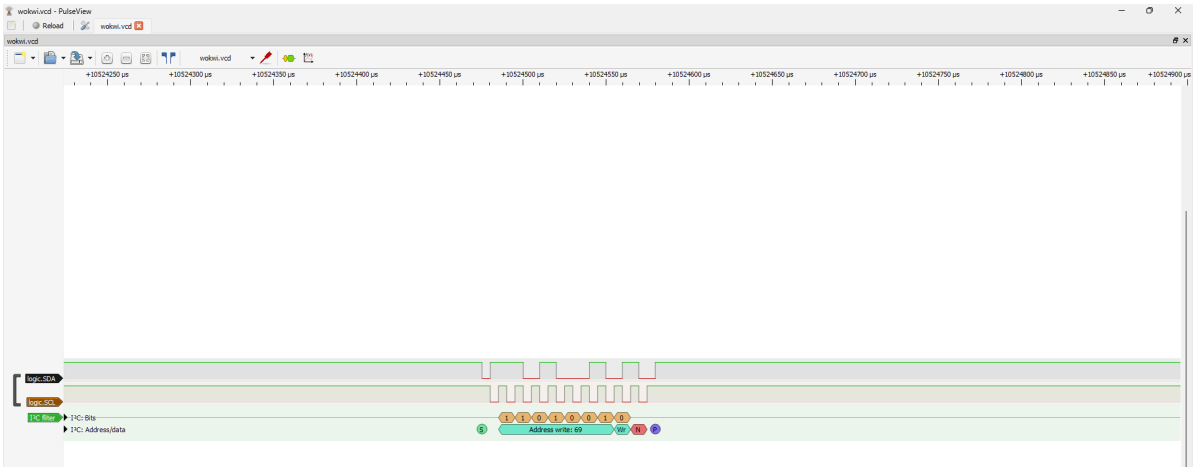


Figure: PulseView trace — 11th transaction